



Department Of Computer Science, CUI Lahore Campus

CSC102 - Discrete Structures

By

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Course Outline

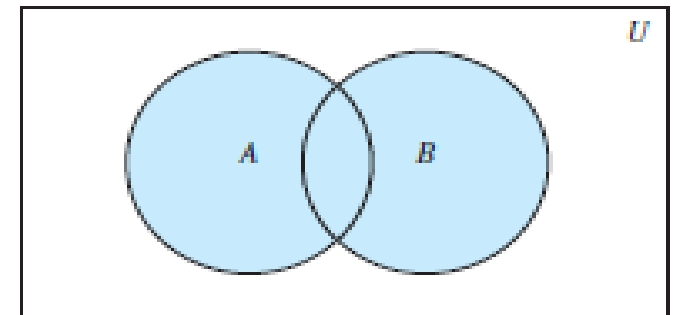
- Sets
 - Set Operations
 - Inclusion-Exclusion Principle of Sets
 - Set Identities
 - Membership Tables

Set Operations

- Two sets can be combined in many different ways.
- Set operations can be used to combine sets.

Union

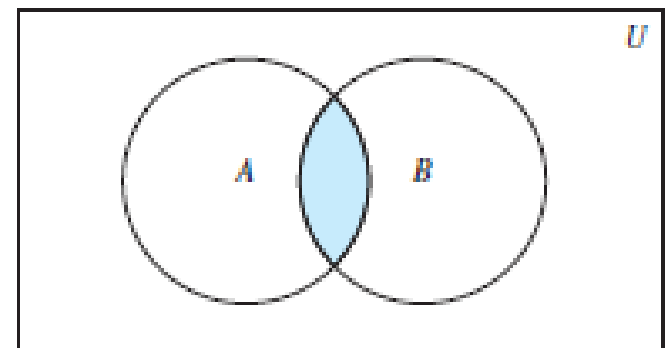
- Let A and B be sets.
- The **union** of A and B , denoted by $A \cup B$, is the set containing those elements that are either in A or in B , or in both.
- $A \cup B = \{x \mid x \in A \vee x \in B\}$



$A \cup B$ is shaded.

Intersection

- Let A and B be sets.
- The **intersection** of A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .
- $A \cap B = \{x \mid x \in A \wedge x \in B\}$



$A \cap B$ is shaded.

Union (example)

- Let $A = \{1, 2, 3\}$
 $B = \{2, 4, 6, 8\}$
 $A \cup B = \{1, 2, 3, 4, 6, 8\}$
- Let $A = \{x \mid x \in \mathbf{Z} \wedge x \text{ is even}\}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$
 $A \cup B = \mathbf{Z}$

Intersection (example)

- Let $A = \{1, 2, 3\}$
 $B = \{2, 4, 6, 8\}$
 $A \cap B = \{2\}$
- Let $A = \mathbf{Z}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$
 $A \cap B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$

Disjoint Sets

- Two sets are called **disjoint** if their intersection is empty.
- Let $A = \{x \mid x \in \mathbf{Z} \wedge x \text{ is even}\}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd}\}$
 $A \cap B = \emptyset$

The Cardinality of the Union of Sets

- **Principle of inclusion-exclusion**
- $|A \cup B| = |A| + |B| - |A \cap B|$

The Cardinality of the Union of Sets

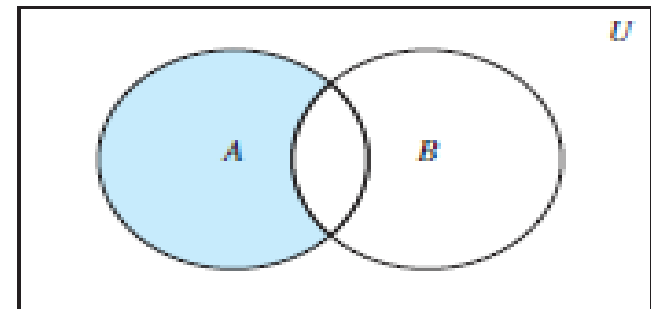
- Find $|A \cup B| = ?$

Solution:

- Let $A = \{1,2,3\}$
 $B = \{2,3,4\}$
 $A \cap B = \{2,3\}$
 $A \cup B = \{1,2,3,4\}$
- $|A| = 3 \quad |B| = 3 \quad |A \cap B| = 2$
- $|A \cup B| = 4$

Difference

- Let A and B be sets.
- The **difference** of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . (also called **complement of B with respect to A**).
- $A - B = \{x \mid x \in A \wedge x \notin B\}$



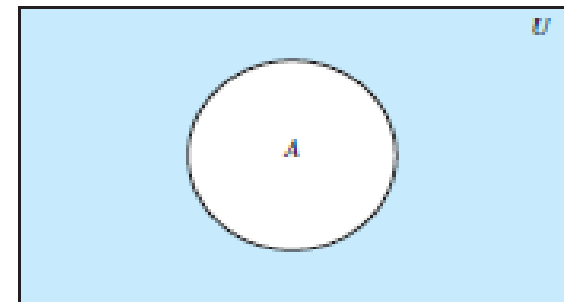
$A - B$ is shaded.

Difference (example)

- Let $A = \{1, 2, 3\}$
 $B = \{2, 4\}$
 $A - B = \{1, 3\}$
- Let $A = \mathbf{Z}$
 $B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is odd} \}$
 $A - B = \{x \mid x \in \mathbf{Z} \wedge x \text{ is even} \}$

Complement

- Let U be the universal set and A be a set.
- The **complement** of A , denoted by $\bar{A}$, is the complement of A with respect to U (which is $U - A$).
- $\bar{A} = \{ x \mid x \notin A \}$



$\bar{A}$ is shaded.

Complement (example)

- Let $A = \{ a, b, c, d \}$ and
 U is the set of English alphabet
 $\bar{A} = \{ e, f, g, h, i, j, k, l, m, n, o, p, q, r, s, t, u, v, w, x, y, z \}$
- Let $A = \{ x \mid x \in \mathbf{Z} \wedge x \text{ is odd} \}$ and
 U is $\mathbf{Z}$
 $\bar{A} = \{ x \mid x \in \mathbf{Z} \wedge x \text{ is even} \}$

Summary Set Operations

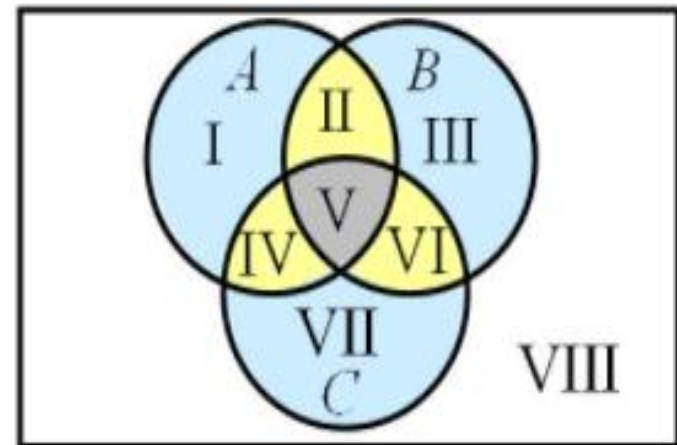
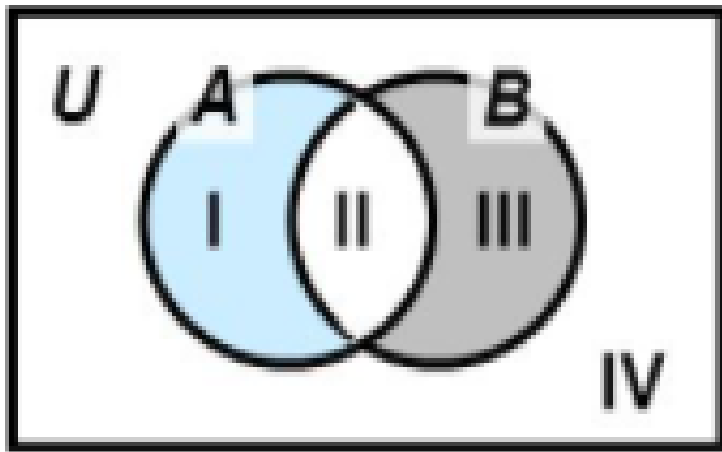
Operation	Notation
Union	$A \cup B = \{x \mid x \in A \vee x \in B\}$
Intersection	$A \cap B = \{x \mid x \in A \wedge x \in B\}$
Difference	$A - B = \{x \mid x \in A \wedge x \notin B\}$
Complement ($U - A$)	$\bar{A} = \{x \mid x \notin A\}$

Inclusion-Exclusion Principle of Sets

2 and 3 - Set Venn Diagram

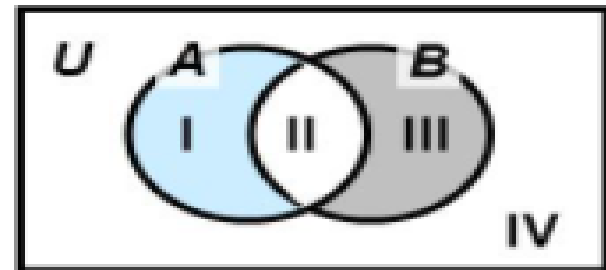
$$|A \cup B| = |A| + |B| - |A \cap B|$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$



2 - Set Venn Diagram

- **Region I** represents the elements in set A that are not in set B.
- **Region II** represents the elements in both sets A and B.
Region III represents the elements in set B that are not in set A.
- **Region IV** represents the elements in the universal set that are in neither set A nor set B.



Example

Suppose a list A contains the 30 students in a mathematics class, and a list B contains the 35 students in an English class, and suppose there are 20 names on both lists.

Find the number of students:

- only on list A
- only on list B
- on list A or B (or both)
- on exactly one list.

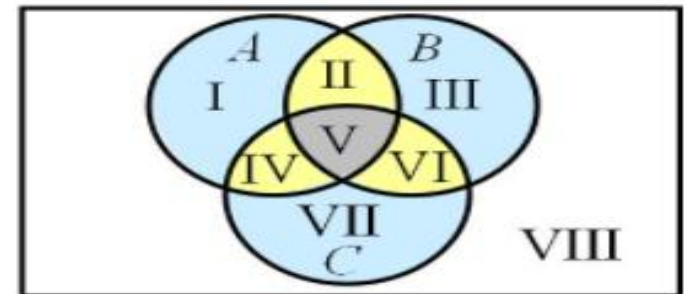
Example

Solution:

- $30 - 20 = 10$ names are only on list A .
- $35 - 20 = 15$ are only on list B .
- $|A \cup B| = |A| + |B| - |A \cap B| = 30 + 35 - 20 = 45$.
- $10 + 15 = 25$ names are only on one list; that is,
 $|A \oplus B| = 25$.

3 - Set Venn Diagram

- **Region I** represents the elements in set A but not in set B or set C.
- **Region II** represents the elements in set A and set B but not in set C.
- **Region III** represents the elements in set B but not in set A or set C.
- **Region IV** represents the elements in sets A and C but not in set B.
- **Region V** represents the elements in sets A, B, and C.
- **Region VI** represents the elements in sets B and C but not in set A.
- **Region VII** represents the elements in set C but not in set A or set B.
- **Region VIII** represents the elements in the universal set U, but not in set A, B, or C.



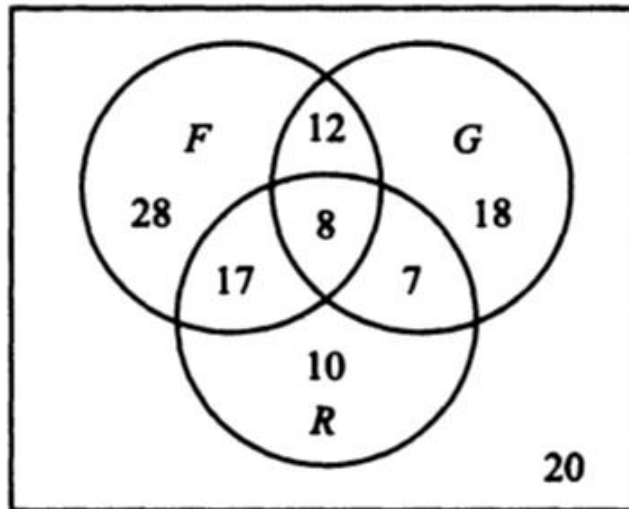
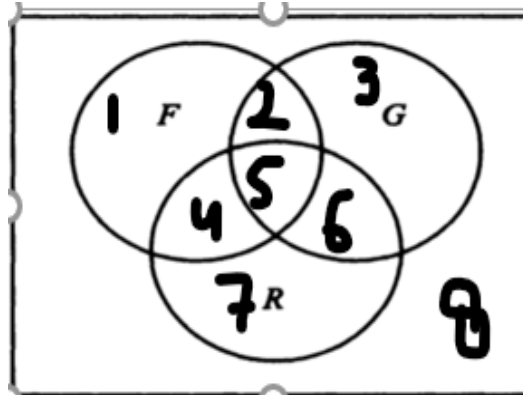
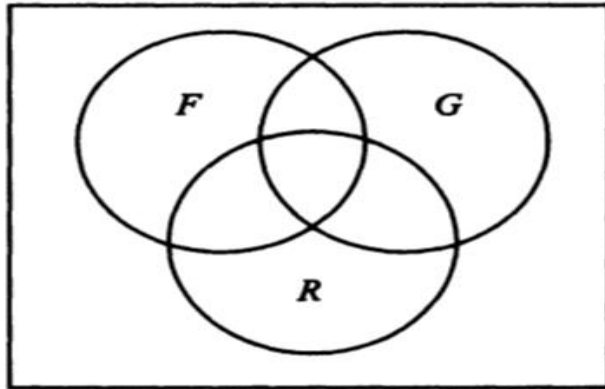
Example

Consider the following data for 120 mathematics students at a college concerning the languages French, German, and Russian:

65 study French, 45 study German,
42 study Russian , 20 study French and German,
25 study French and Russian,
15 study German and Russian.
8 study all three languages.

Determine how many students study exactly 1 language course and fill the correct numbers of students in each eight region of Venn diagram shown in figure.

Example



$$U = 120$$

$$F = 65$$

$$G = 45$$

$$R = 42$$

$$F \cap G = 20$$

$$F \cap R = 25$$

$$G \cap R = 15$$

$$F \cap G \cap R = 8$$

$$R1 - |F| = 65 - 12 - 8 - 17 = 28$$

$$R3 - |G| = 45 - 12 - 8 - 7 = 18$$

$$R7 - |R| = 42 - 17 - 8 - 7 = 10$$

$$R2 - |F \cap G| = 20 - 8 = 12$$

$$R4 - |F \cap R| = 25 - 8 = 17$$

$$R6 - |G \cap R| = 15 - 8 = 7$$

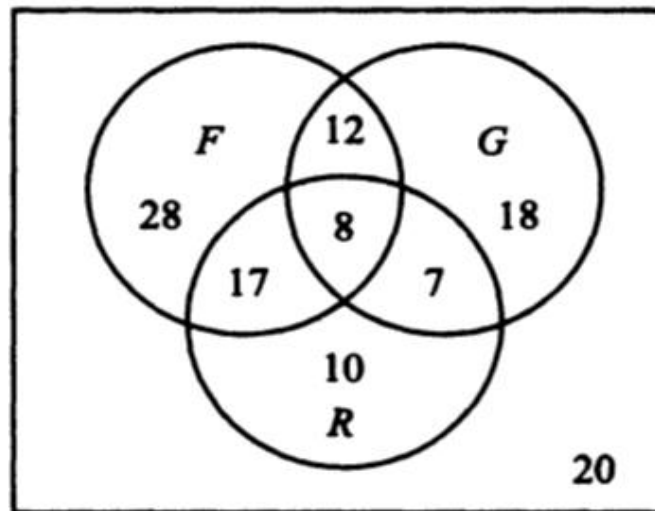
$$R5 - |F \cap G \cap R| = 8$$

$$|F \cup G \cup R| = 65 + 45 + 42 - 20 - 25 - 15 + 8 = 100$$

$$R8 - |U| - |F \cup G \cup R| = 120 - 100 = 20$$

Example

- Total number of students exactly registered in one course
= $28+18+10=56$



Set Identities

$$A \cup \emptyset = A$$

$$A \cap U = A$$

Identity Laws

$$A \cup U = U$$

$$A \cap \emptyset = \emptyset$$

Domination Laws

$$A \cup A = A$$

$$A \cap A = A$$

Idempotent Laws

$$\overline{(\overline{A})} = A$$

Complementation Law

$$A \cup \overline{A} = U$$

$$A \cap \overline{A} = \emptyset$$

Complement Laws

Set Identities

$$A \cup B = B \cup A$$

$$A \cap B = B \cap A$$

Commutative Laws

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

Associative Laws

$$A \cup (A \cap B) = A$$

$$A \cap (A \cup B) = A$$

Absorption Laws

Set Identities

$$\overline{A \cup B} = \bar{A} \cap \bar{B}$$

De Morgan's Law

$$\overline{A \cap B} = \bar{A} \cup \bar{B}$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Distributive Law

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A - B = A \cap \bar{B}$$

Difference Law

How to Prove a Set Identity

- Four methods:
 - Use membership tables
 - Use the basic set identities
 - Prove each set is a subset of each other
 - Use set builder notation and logical equivalences

What is a membership table

- Membership tables show all the combinations of sets an element can belong to
 - 1 means the element belongs, 0 means it does not
- Consider the following membership table:

A	B	$A \cup B$	$A \cap B$	$A - B$
1	1	1	1	0
1	0	1	0	1
0	1	1	0	0
0	0	0	0	0

Membership Table

A	B	B-A	$\overline{A}$	$\overline{B}$	$\overline{A \cap B}$	$\overline{A} \cup \overline{B}$
1	1	0	0	0	0	0
1	0	0	0	1	1	1
0	1	1	1	0	1	1
0	0	0	1	1	1	1

Membership Table

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

Distributive Law

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

Set Identities (example)

- Show $\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$

Set Identities (example)

- Show $\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$

Solution:

$$\begin{aligned}\overline{A \cup (B \cap C)} &= \bar{A} \cap \overline{(B \cap C)} && \text{(By DeMorgan's Law)} \\ &= \bar{A} \cap (\bar{B} \cup \bar{C}) && \text{(By DeMorgan's Law)} \\ &= \bar{A} \cap (\bar{C} \cup \bar{B}) && \text{(By Commutative Law)} \\ &= (\bar{C} \cup \bar{B}) \cap \bar{A} && \text{(By Commutative Law)}\end{aligned}$$

Set Identities

$$A \cup (B - A) = A \cup B$$

$$L.H.S = A \cup (B - A)$$

$$= A \cup (B \cap \bar{A})$$

$$= (A \cup B) \cap (A \cup \bar{A})$$

$$= (A \cup B) \cap U$$

$$= A \cup B$$

$$= R.H.S$$

Difference Law

Distribution Law

Complement Law

Identity Law

Exercise Questions

Chapter # 2

Topic # 2.2

Question # 1, 2, 3,4,15 - b,16 –(d,e),19-b,20 – (d,e),
21, 26, 27